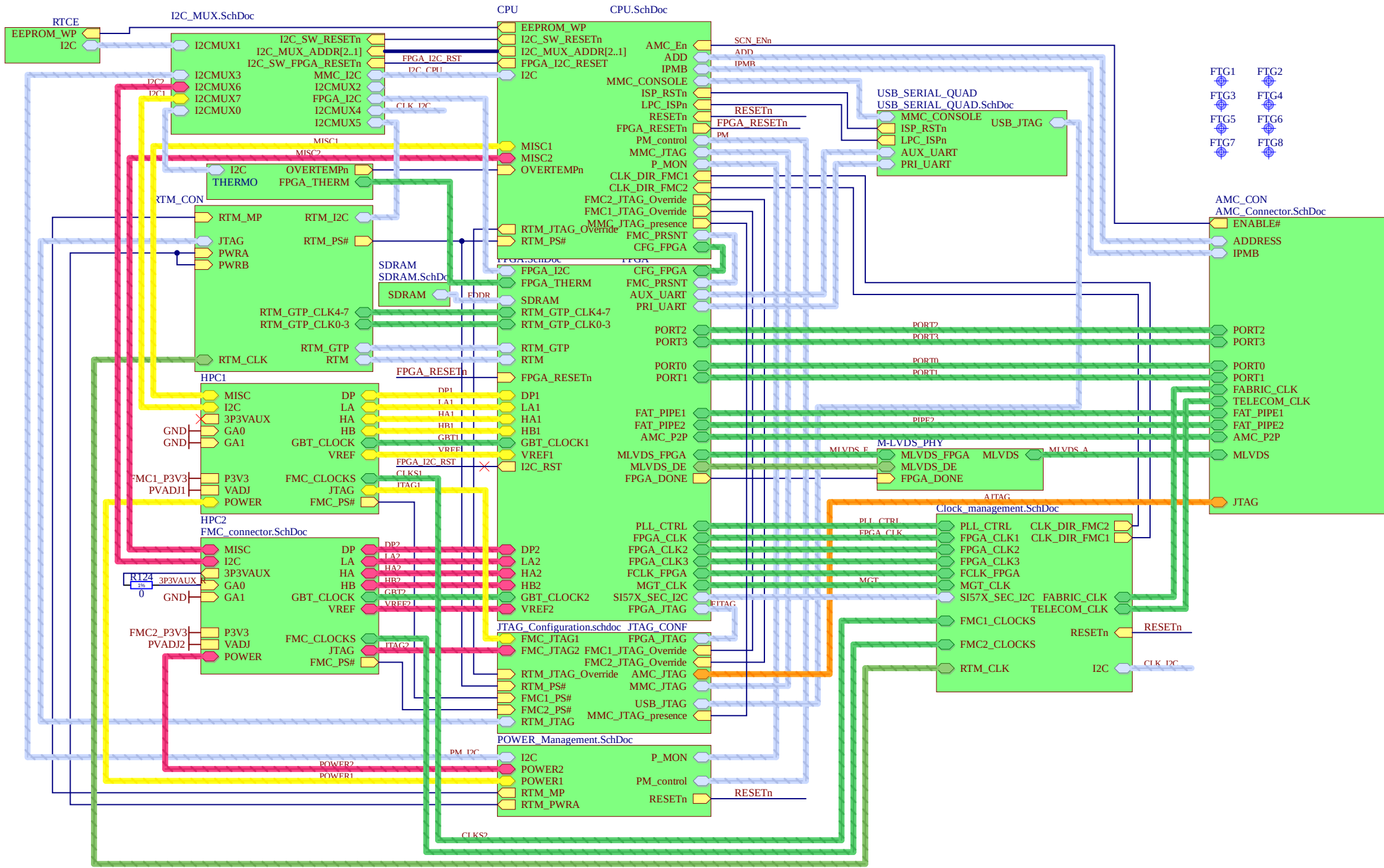
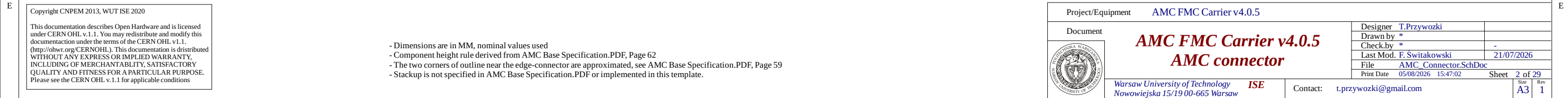


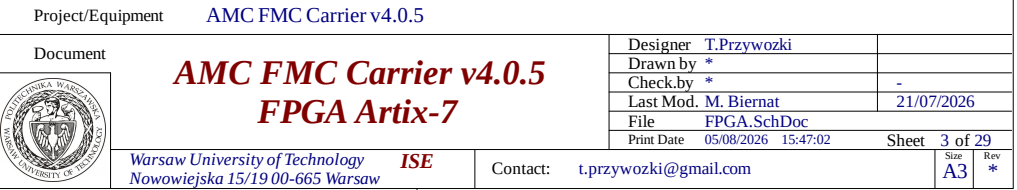


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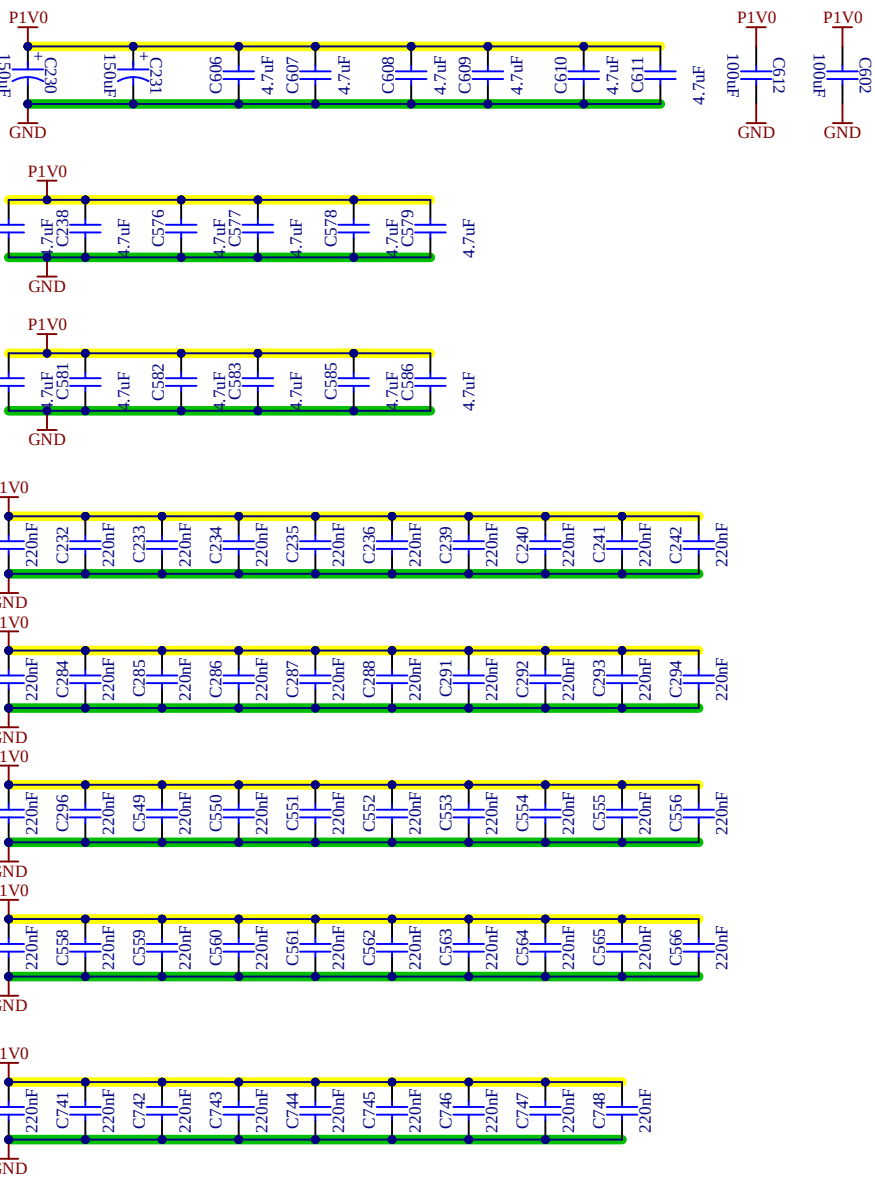
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	Check by	*	-		
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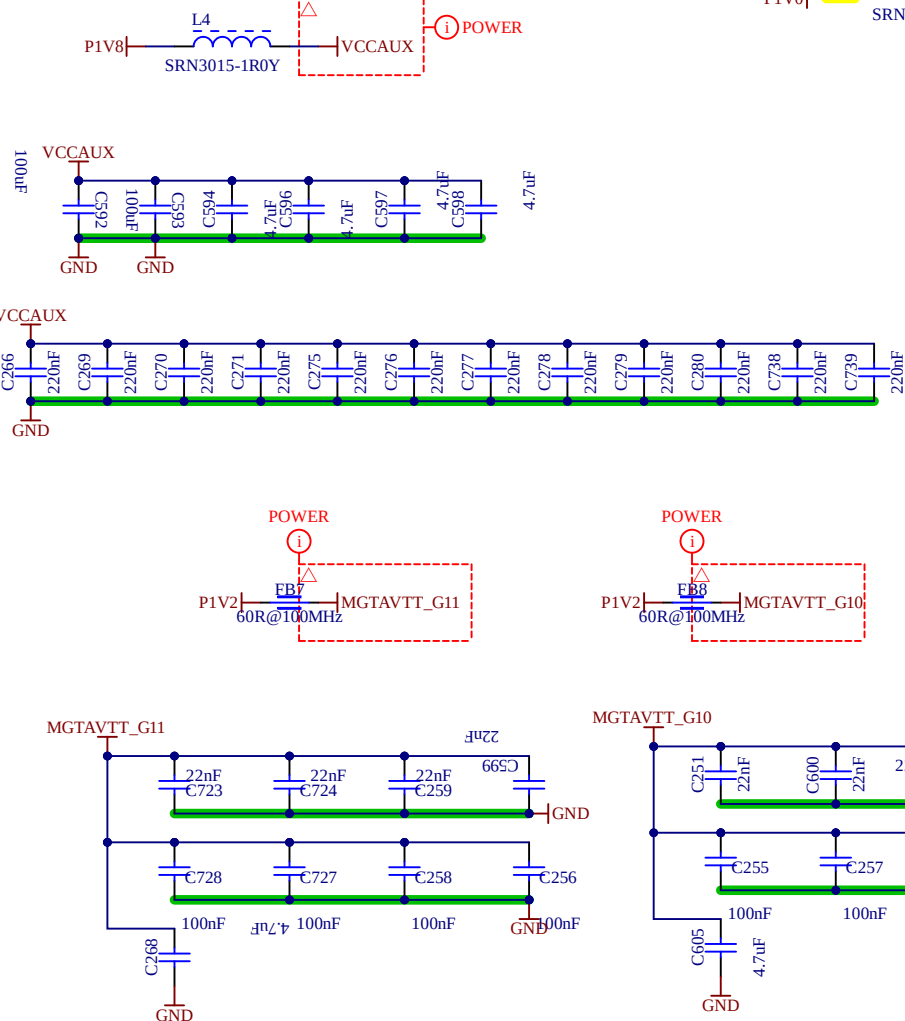
VCCINT

1x680uF, 2x100uF, 28x4.7uF
42x0.47uF on PG.34



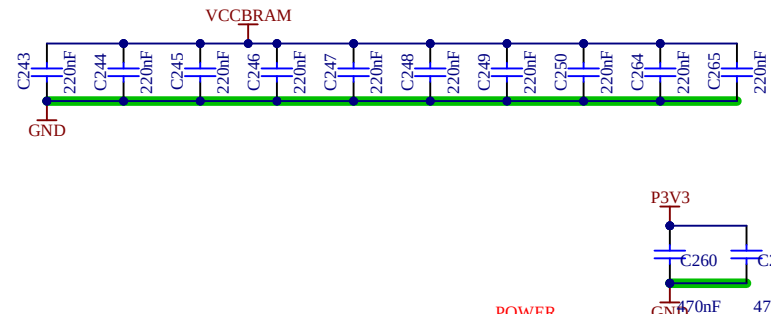
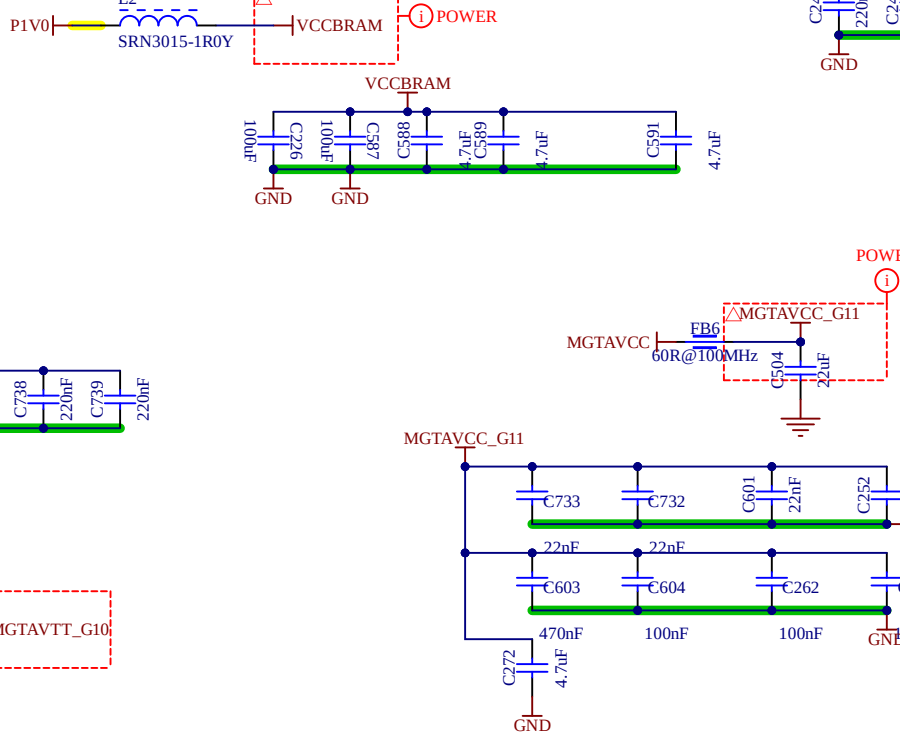
VCCAUX

3x47uF, 4x4.7uF, 7x0.47uF



VCCBRAM

2x100uF, 4x4.7uF, 8x0.47uF



MGTAVCC	1V	2.3A		
MGTAVTT	1.2V	1.5A		
MGTAVCCAUX	1.8V	0.5A		
VCCAUX	1.8V	1A	seq1	
VCCAUX_JO	1.8/2V	0.1A	seq1	
VCCBRAM	1V	1.8A	seq0	
VCCINT	1V	6A	seq0	
VCCO_12	1.2V	3.2A	seq2	
VCCO_135	1.35V	0.9A	seq2	
VCCO_15	1.5V	0.9A	seq2	
VCCO_18	1.8V	0.9A	seq2	
VCCO_25	2.5V	0.9A	seq2	
VCCO_33	3.3V	0.9A	seq2	

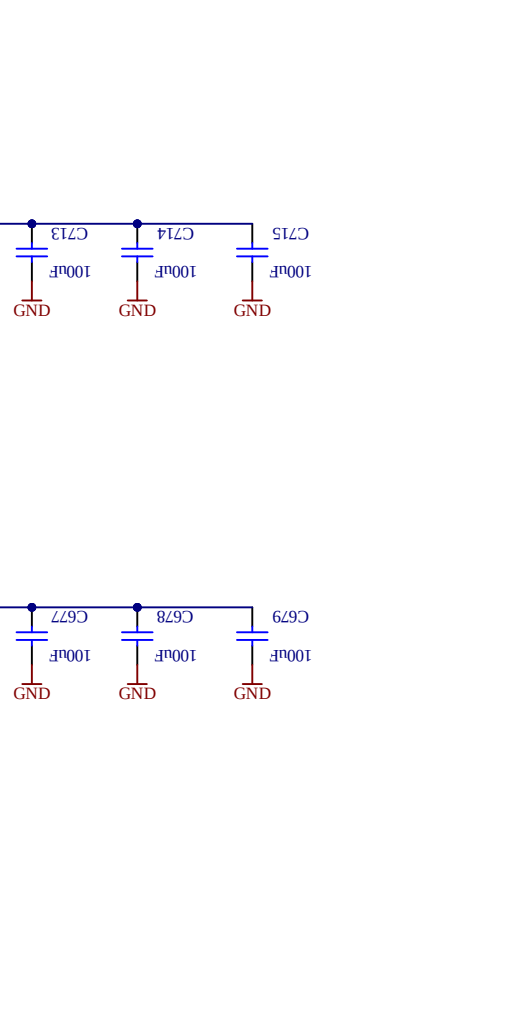
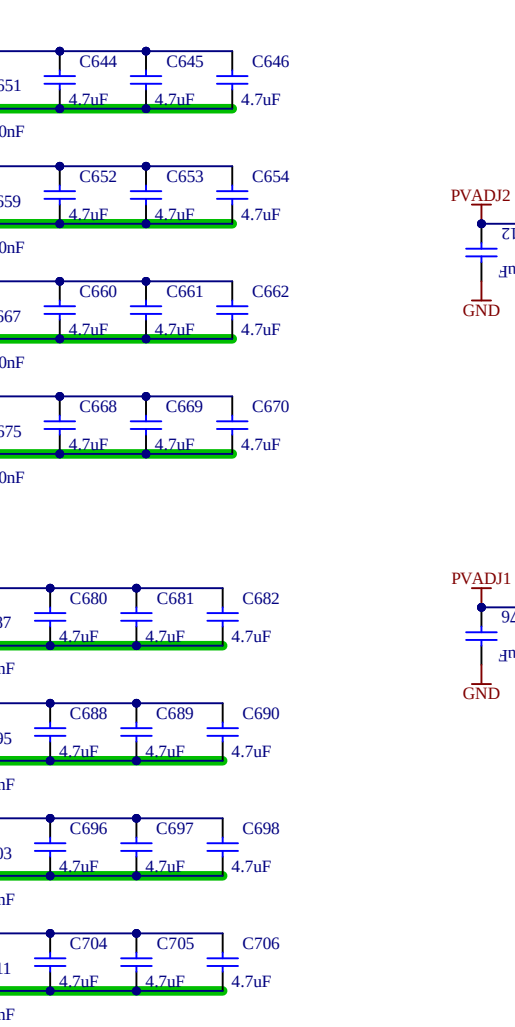
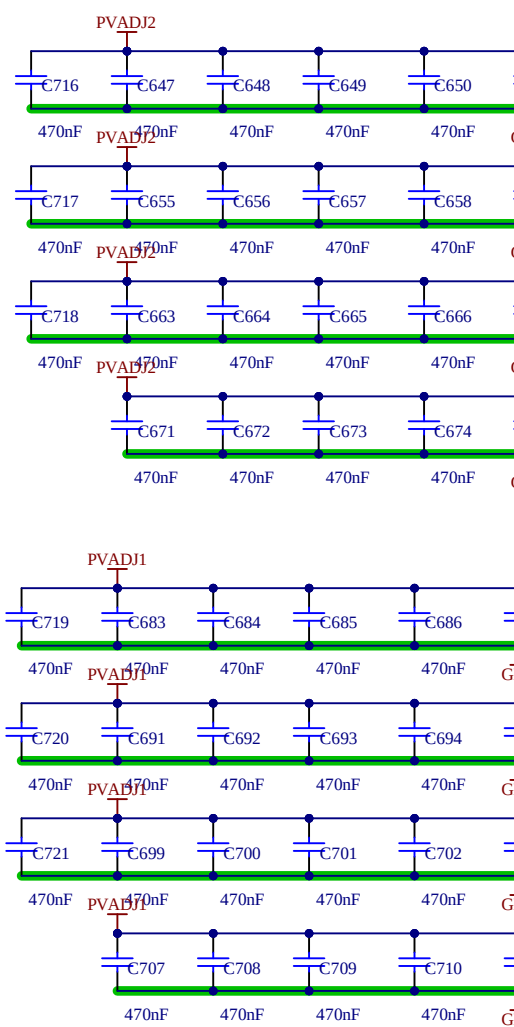
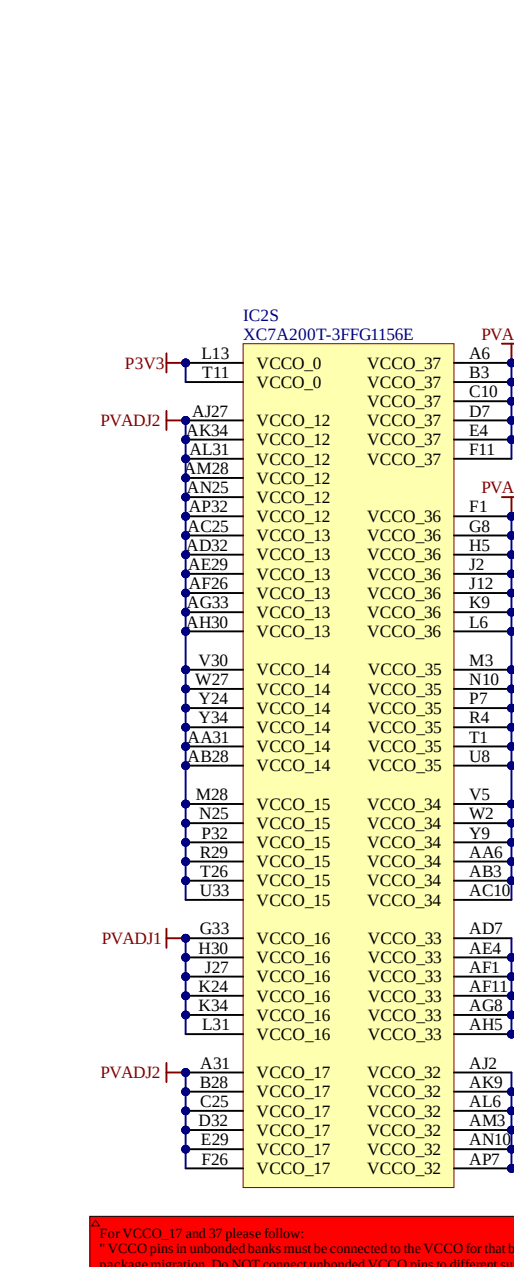
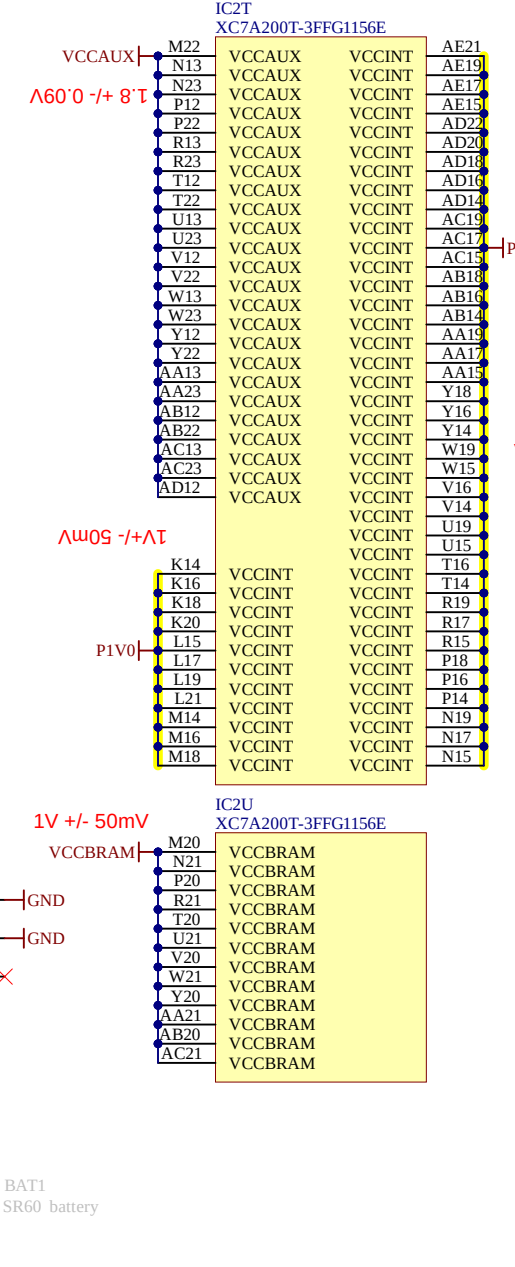
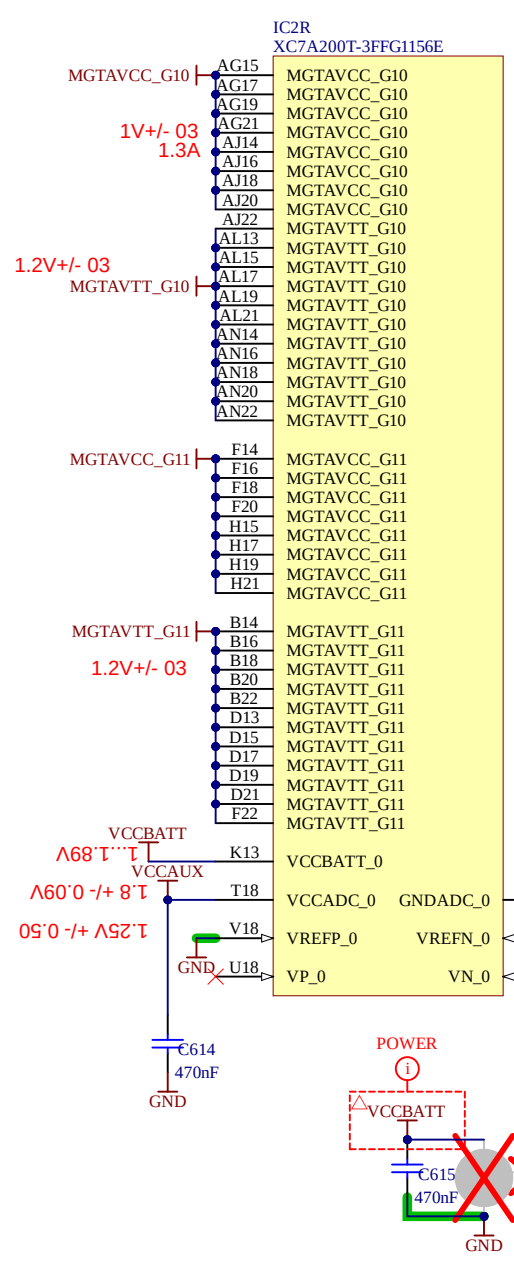
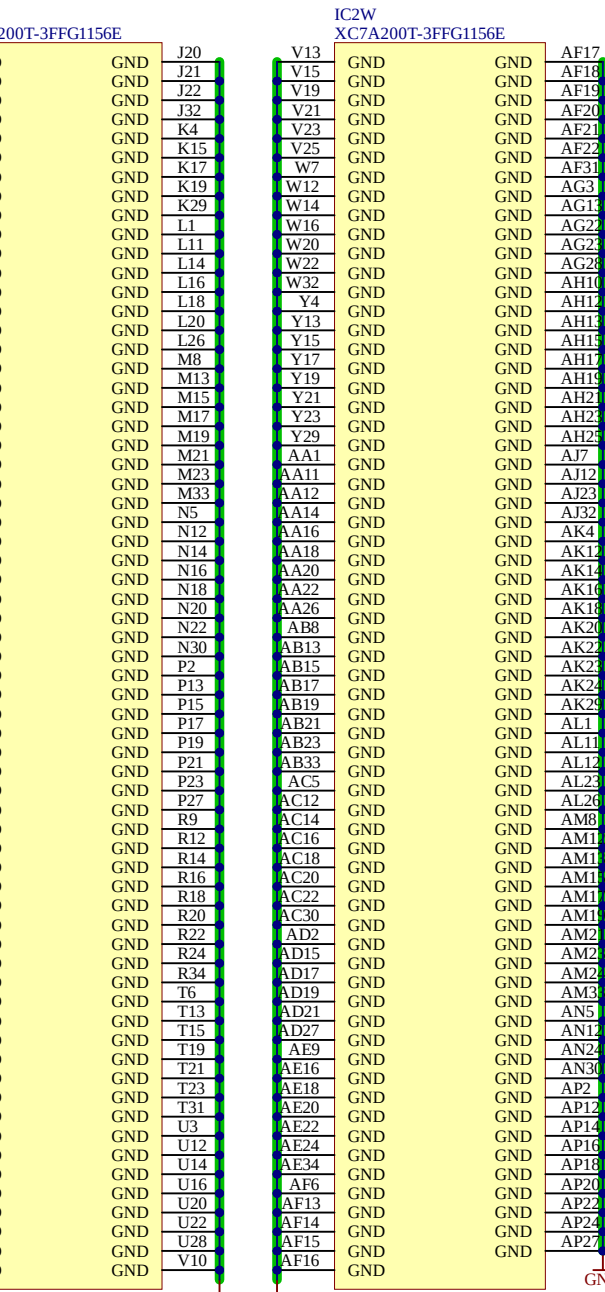
power-on sequence is VCCINT, VCCBRAM, VCCAUX, and VCCO voltage difference between VCCO and VCCAUX must not exceed 2.625V for longer than TVCCO2VCCAUX

power-on sequence to achieve minimum current draw for the GTP transceivers is VCCINT, VMGTAVCC.

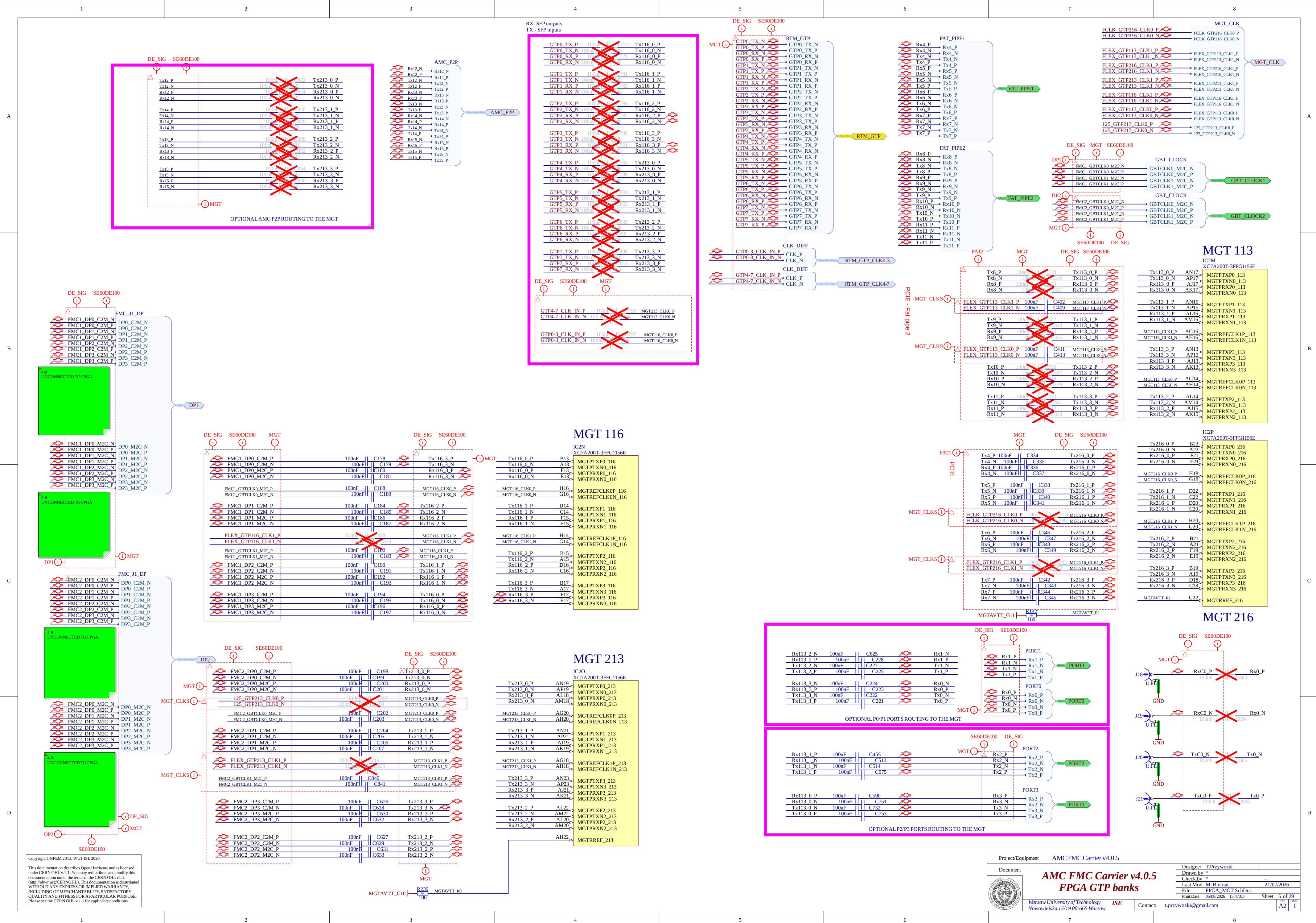
VMGTAVTT OR VMGTAVCC, VCCINT, VMGTAVTT. There is no recommended sequencing for VMGTAVCCAUX. Both VMGTAVCC and VCCINT can be ramped simultaneously. The recommended power-off sequence is the reverse of the power-on sequence to achieve minimum current draw.

When VMGTAVTT is powered before VMGTAVCC and VMGTAVTT - VMGTAVCC > 150 mV and VMGTAVCC < 0.7V, the VMGTAVTT current draw can increase by 460 mA per transceiver during VMGTAVCC ramp up. The duration of the current draw can be up to 0.3 x TMGTAVCC (ramp time from GND to 90% of VMGTAVCC). The reverse is true for power-down.

• When VMGTAVTT is powered before VCCINT and VMGTAVTT - VCCINT > 150 mV and VCCINT < 0.7V, the VMGTAVTT current draw can increase by 50 mA per transceiver during VCCINT ramp up. The duration of the current draw can be up to 0.3 x TVCCINT (ramp time from GND to 90% of VCCINT). The reverse is true for power-down.



For VCCO_17 and 37 please follow:
* VCCO pins in unbonded banks must be connected to the VCCO for that bank for package migration. Do NOT connect unbonded VCCO pins to different supplies. Without package migration requirement, VCCO pins in unbonded banks can be tied to a common supply (VCCO or ground).
No migration available (since there is no 350T) - I recommend connecting to empty power rail and using as power decoupling placeholder for decaps. Remove bulks C222 and C219.



A

A

B

B

C

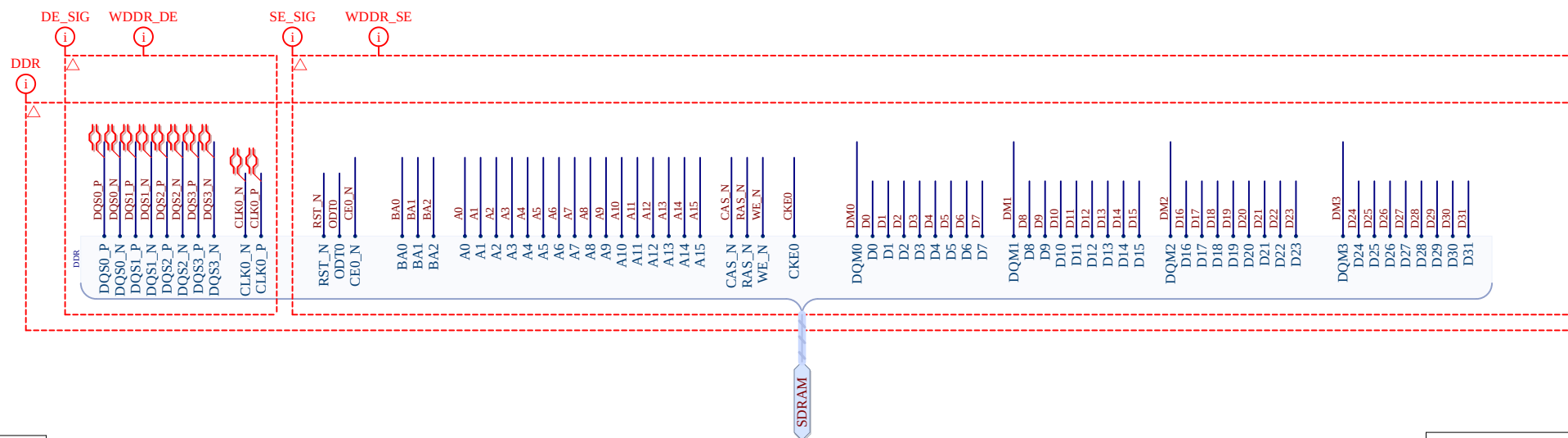
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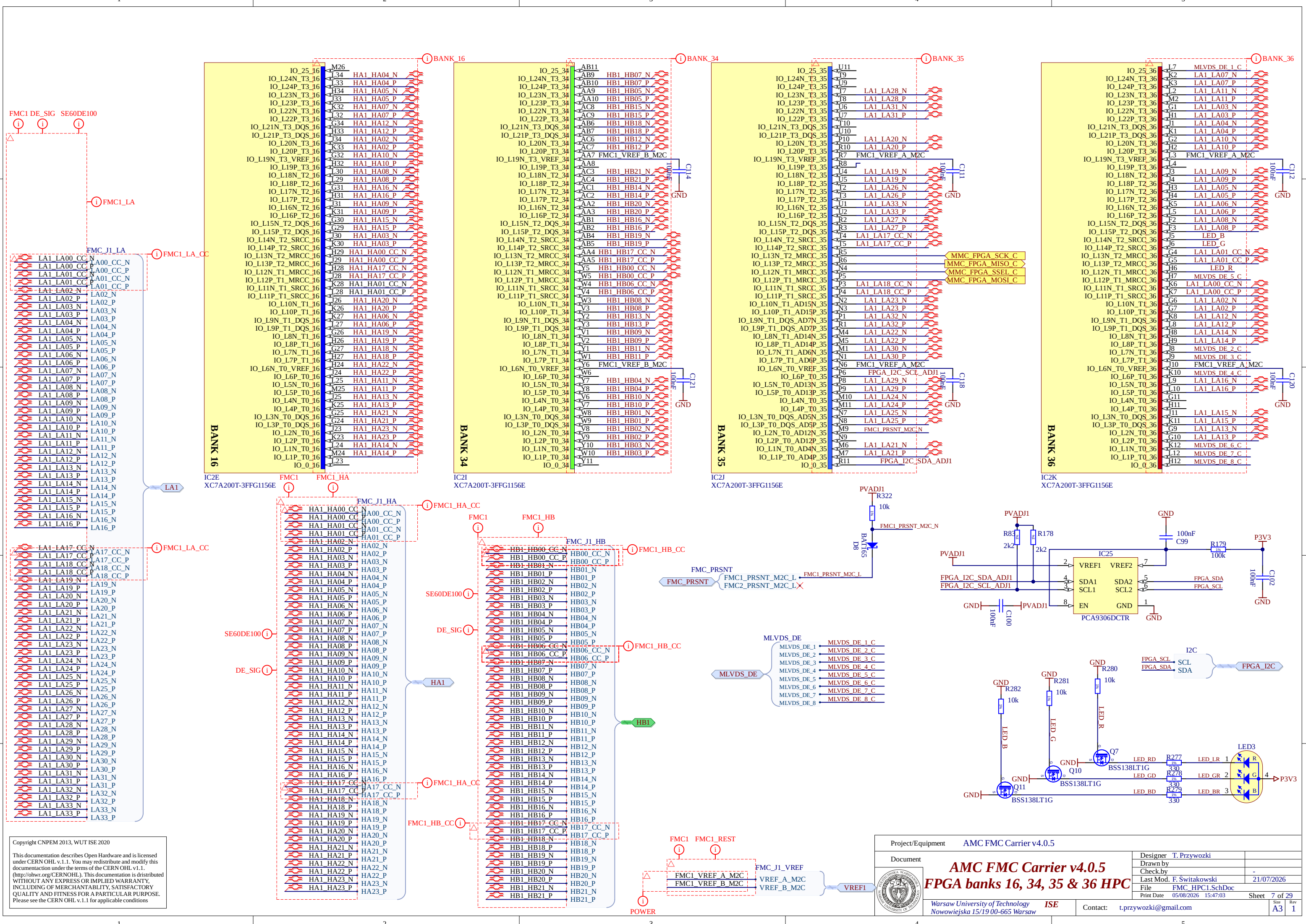
D

D

E

E





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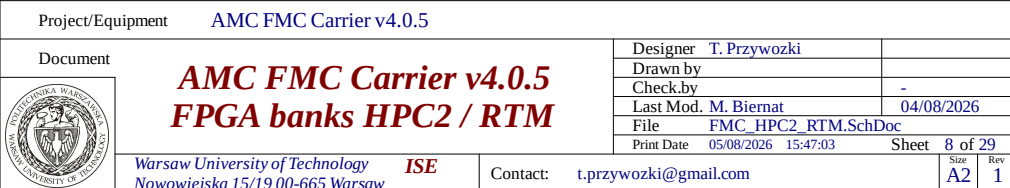
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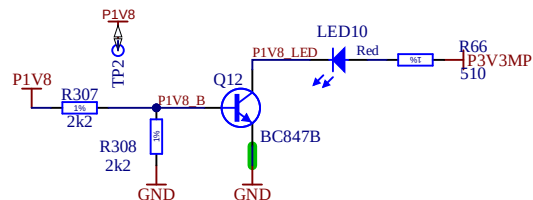
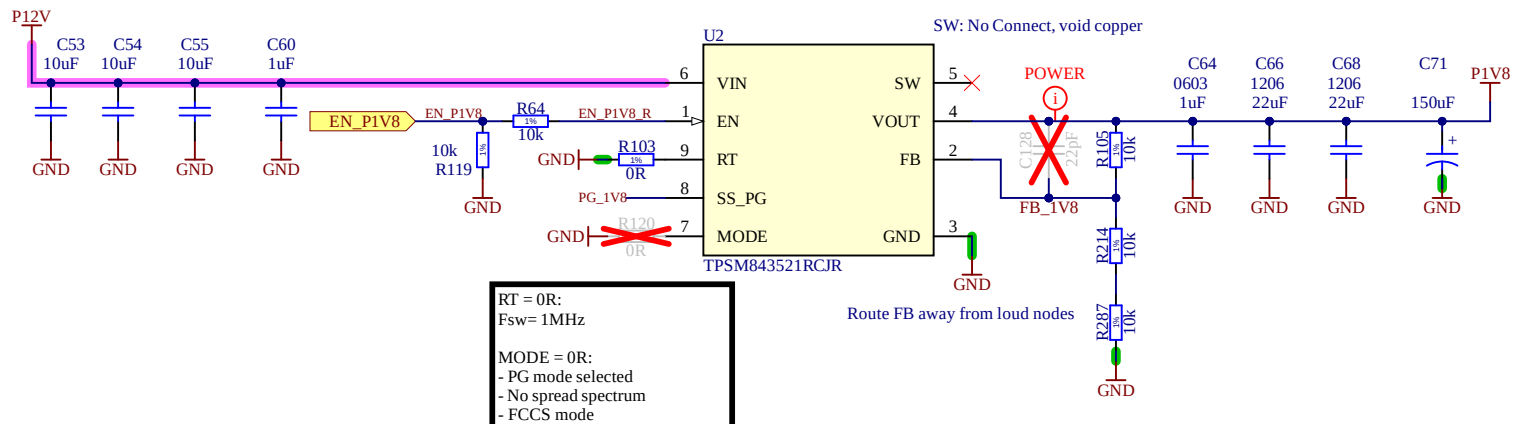
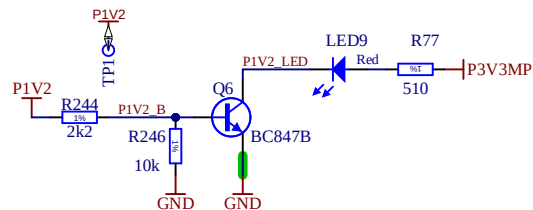
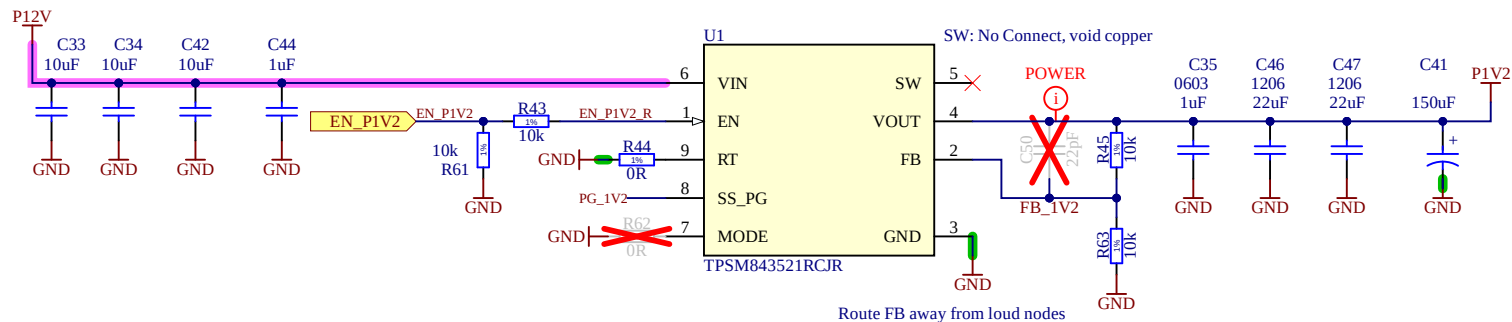
AMC FMC Carrier v4.0.5
FPGA banks 16, 34, 35 & 36 HPC

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Compatible parts:
TPSM843521 - 5A
TPSM843321 - 3A
TPSM84538 - 5A, 28V in
TPSM84338 - 3A, 28V in

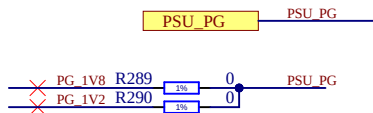


RT = 0R:
Fsw = 1MHz

MODE = 0R:
- PG mode selected
- No spread spectrum
- FCCS mode

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Document

AMC FMC Carrier v4.0.5
Power P1V2 & P1V8



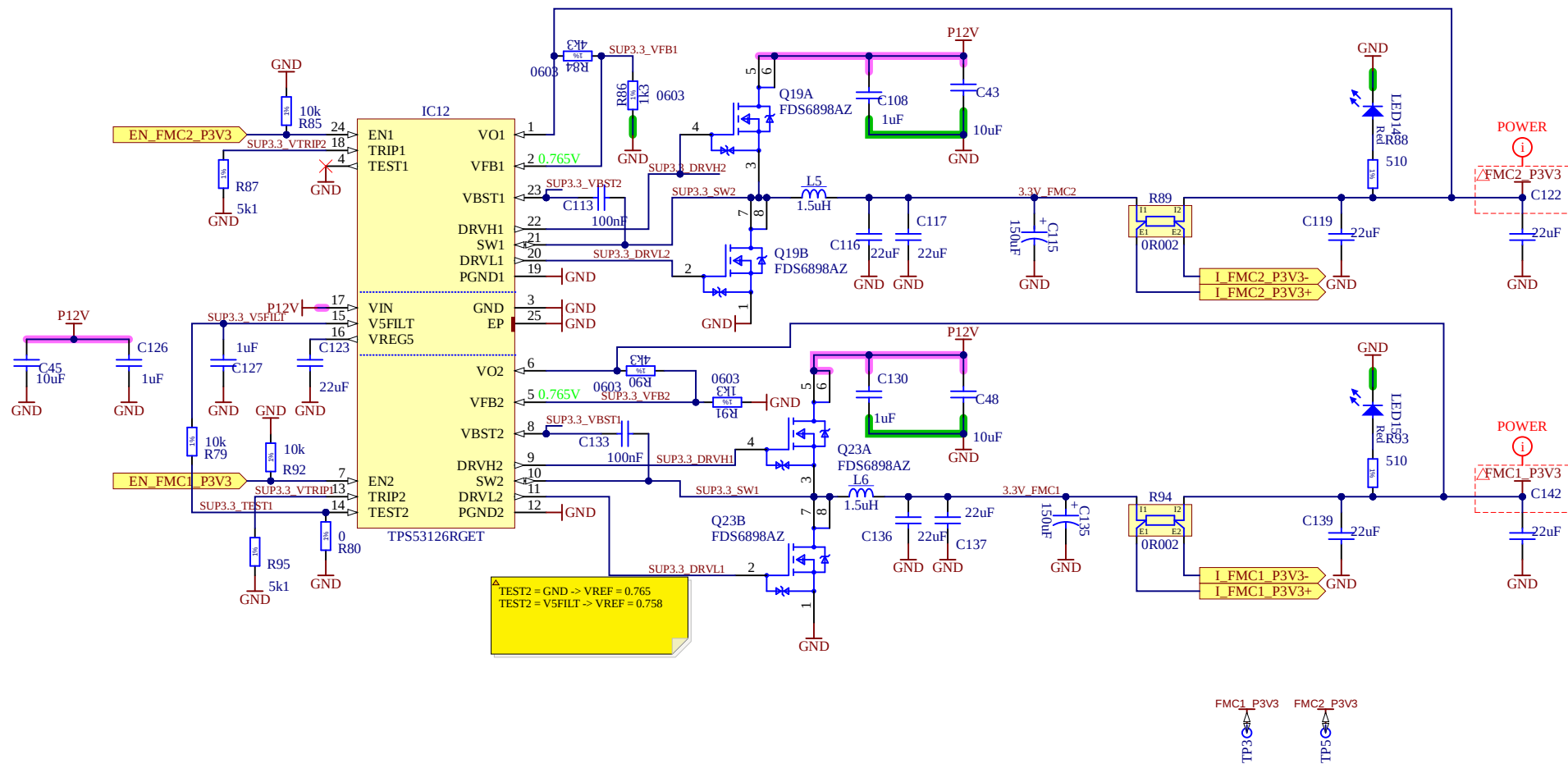
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Project/Equipment AMC FMC Carrier v4.0.5

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AMC FMC Carrier v4.0.5 Power FMCs P3V3



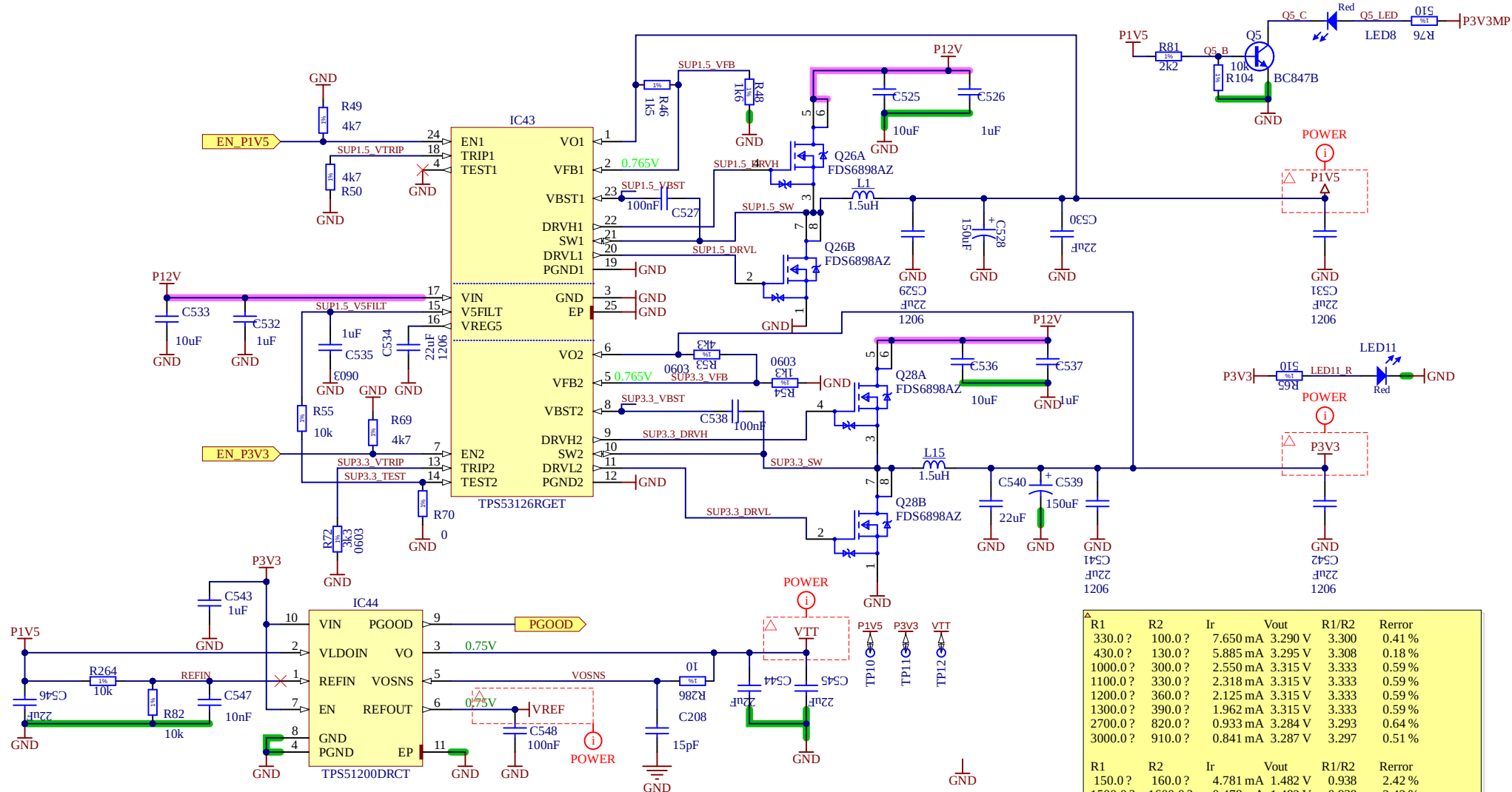
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Document



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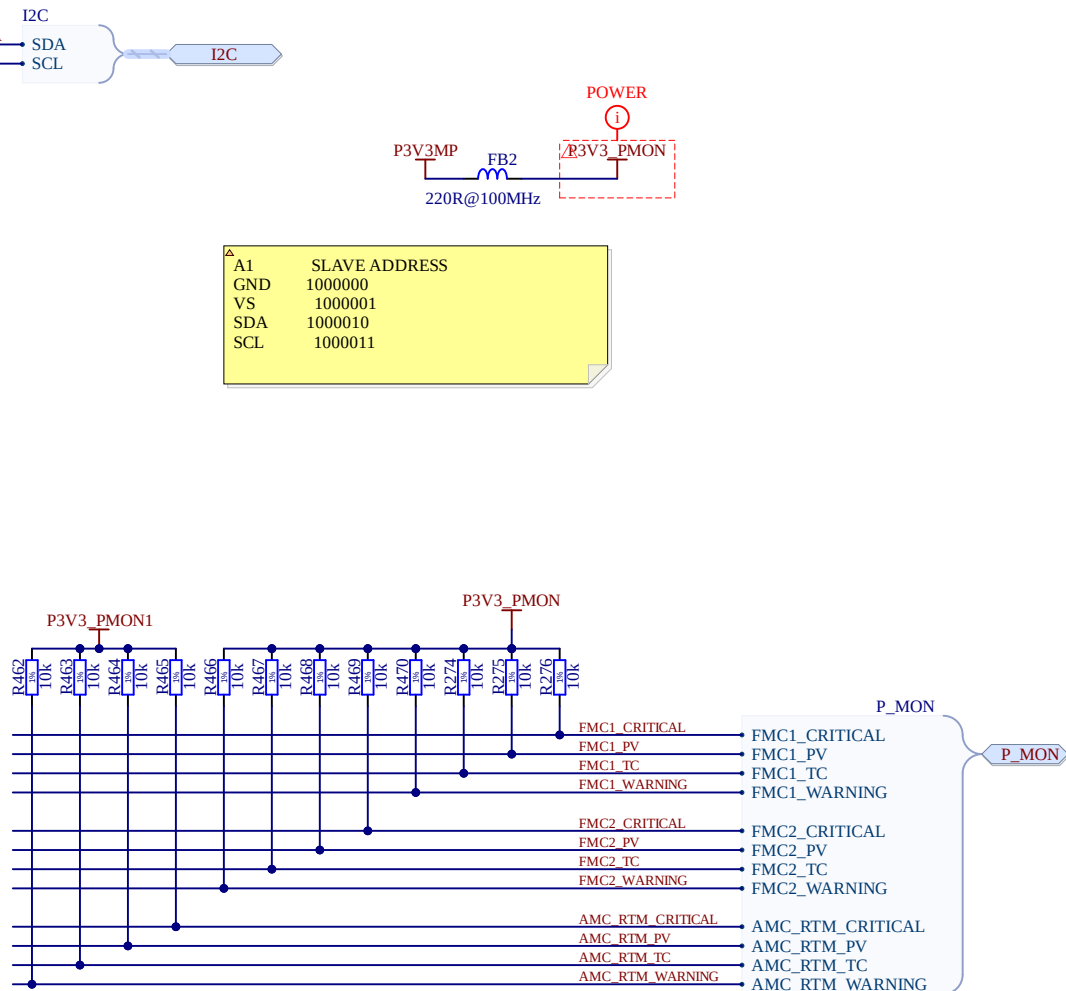
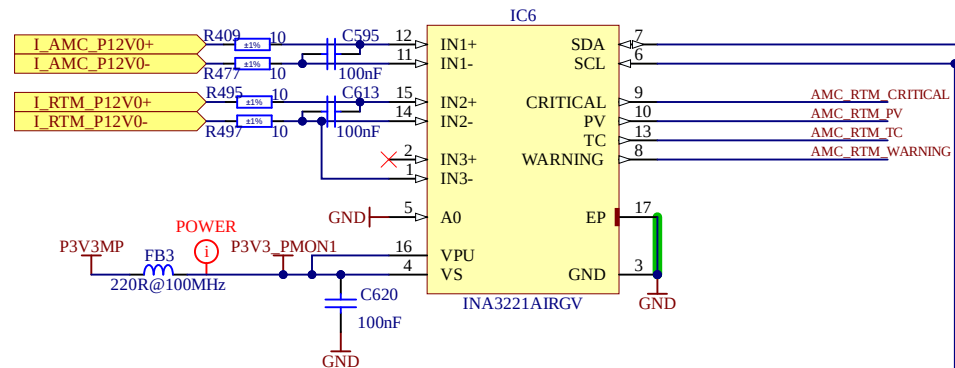
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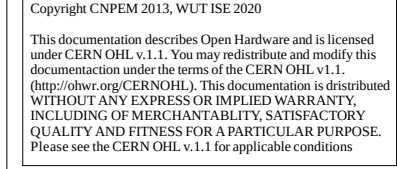
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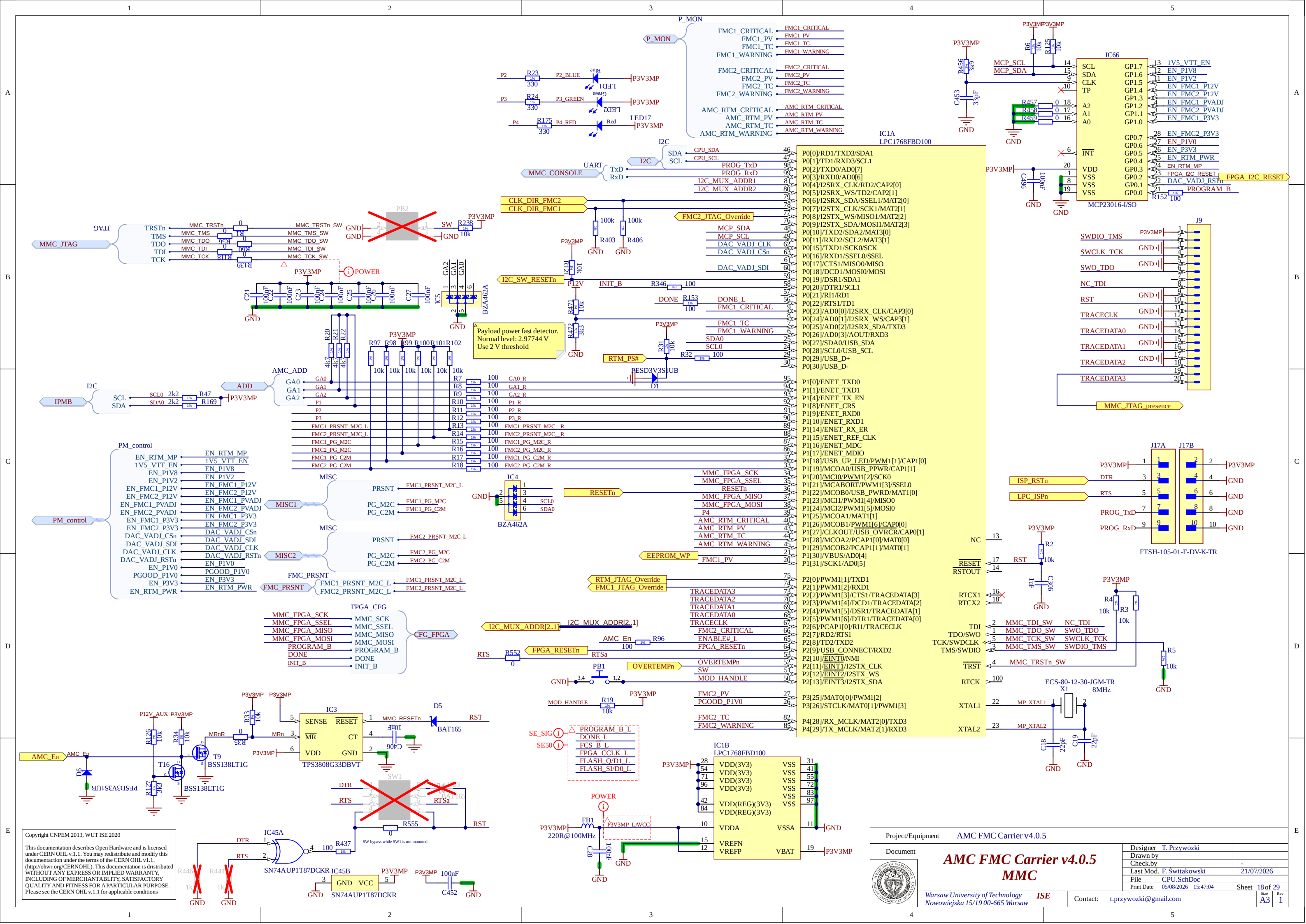
Warsaw University of Technology
Nowowiejska 15/19 00-665 Warsaw

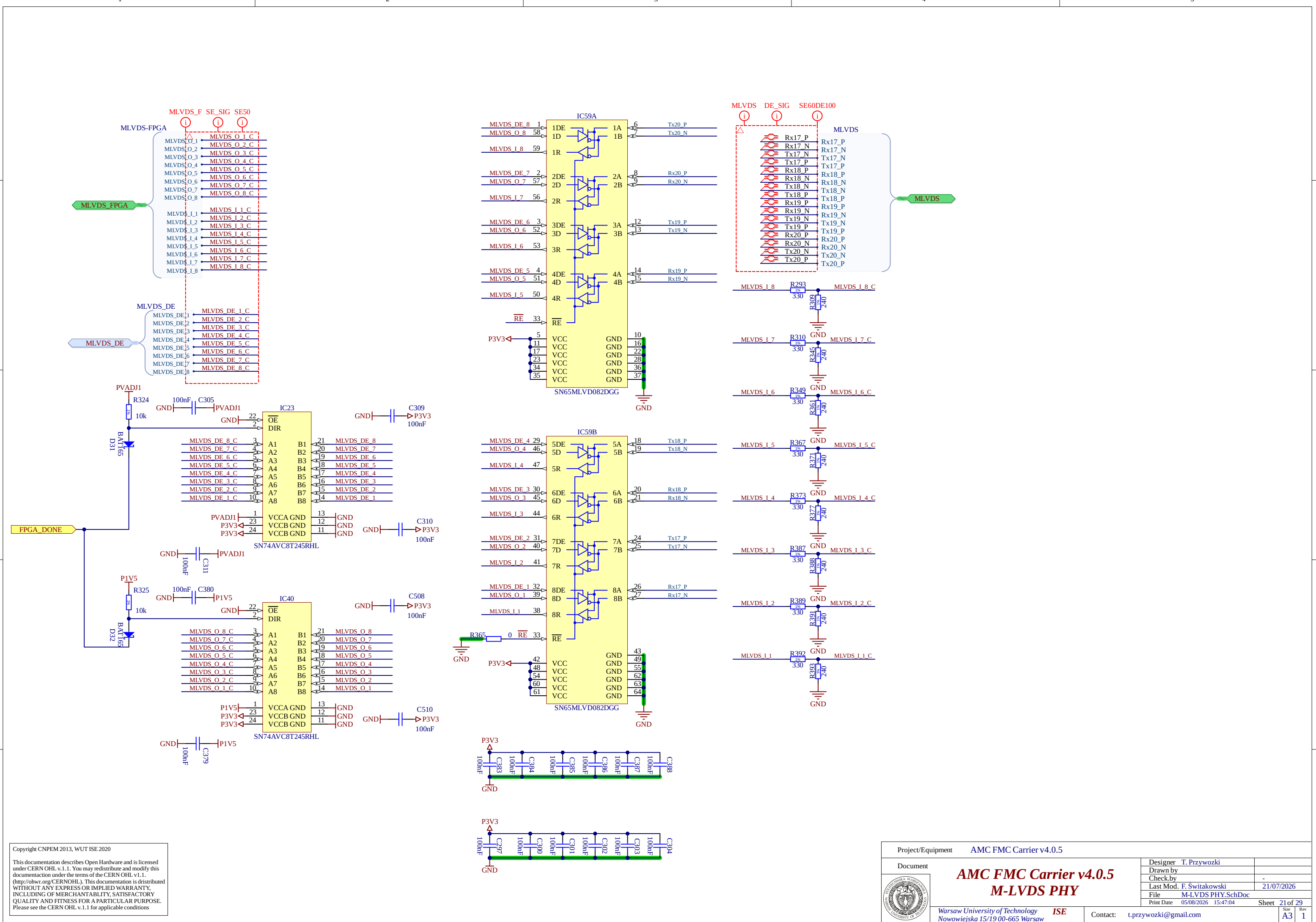
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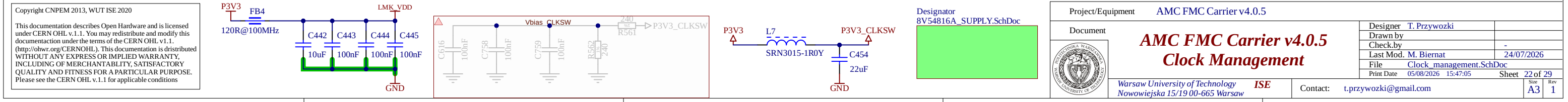


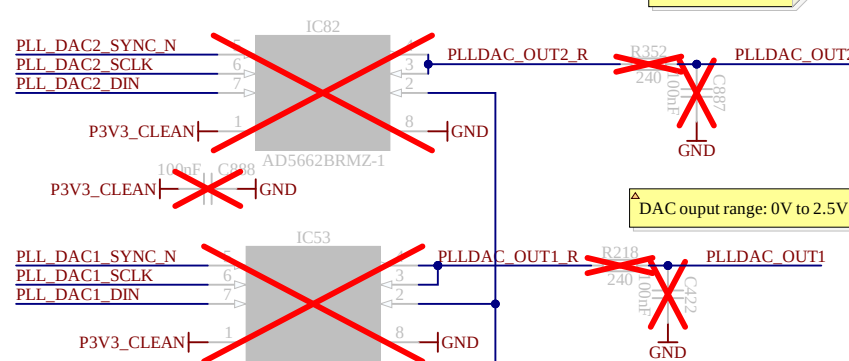
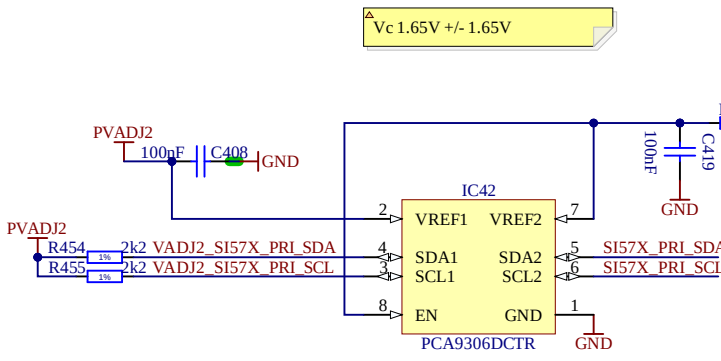
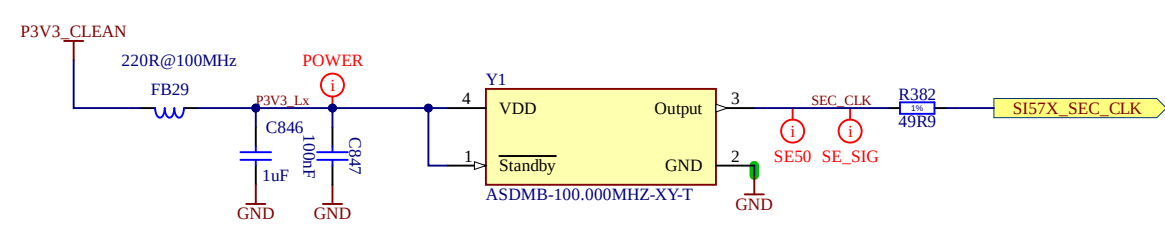
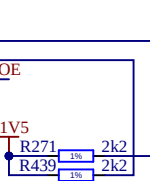
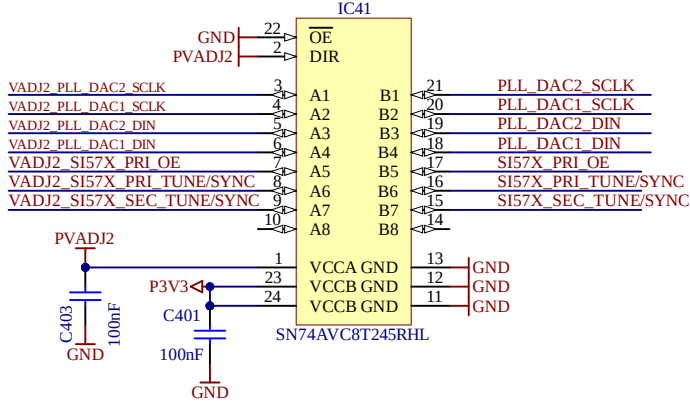
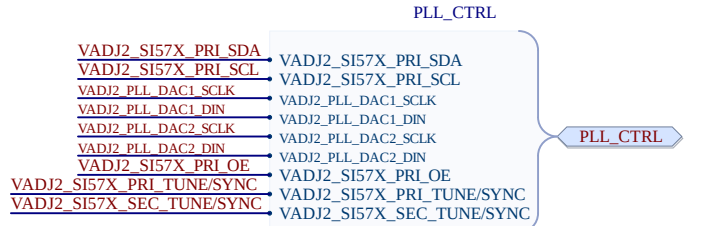




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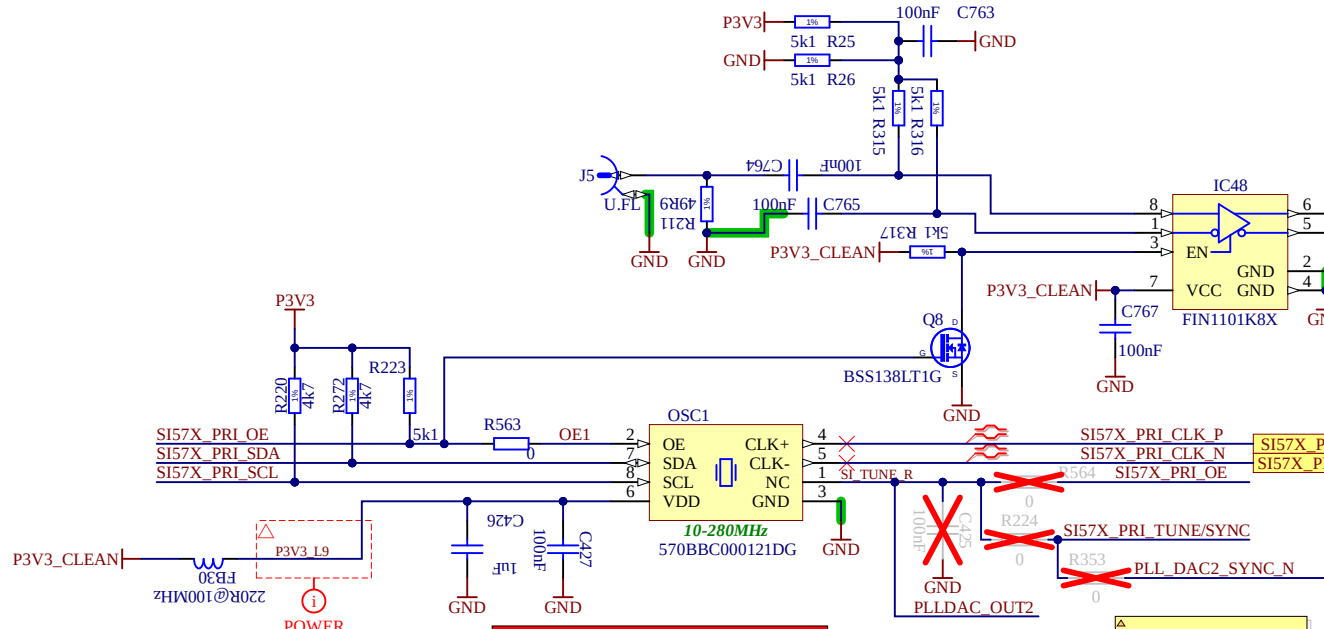
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DAC Vih = 2V so it accepts 2.5V CMOS signal

DAC output range: 0V to 2.5V



oscillators available in variants

549BACB001937ABG
571BFB000449DG
570BBC000121DG

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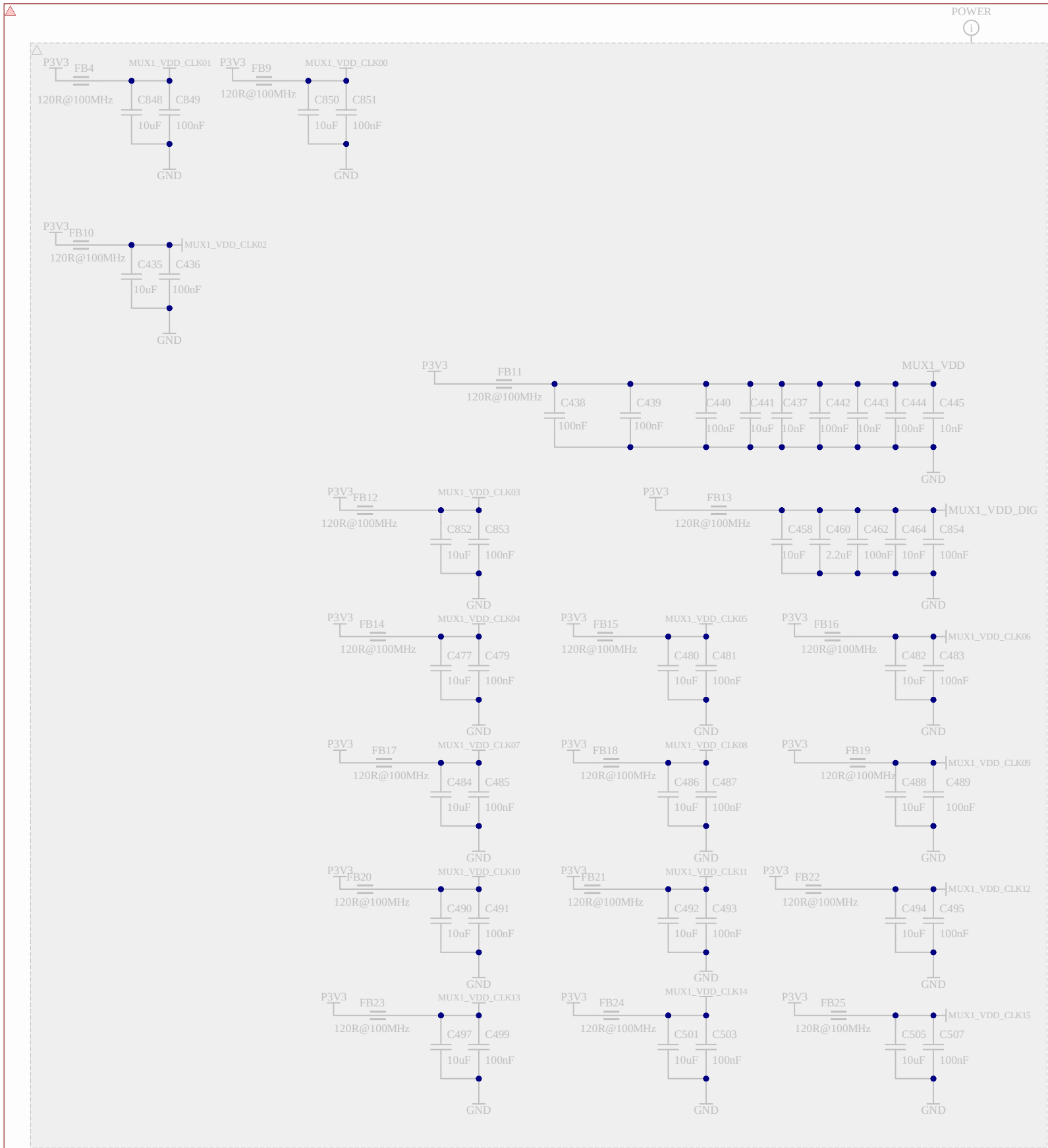
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AMC FMC Carrier v4.0.5 Oscillators

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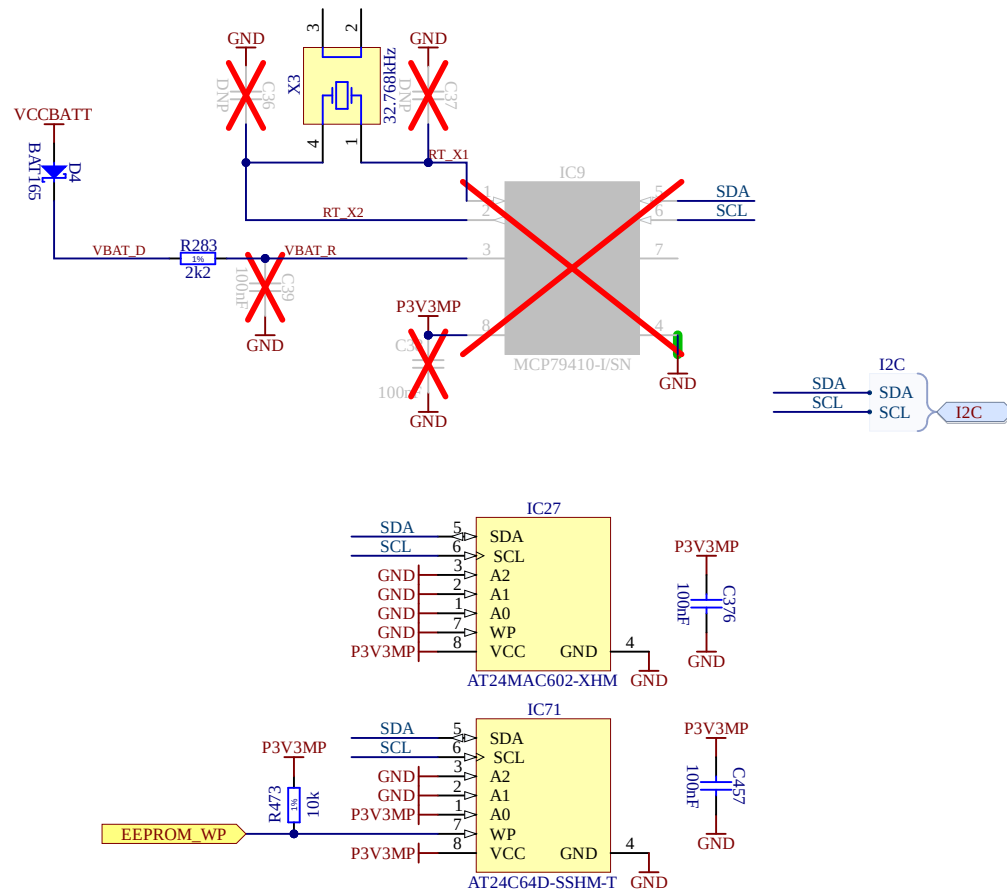
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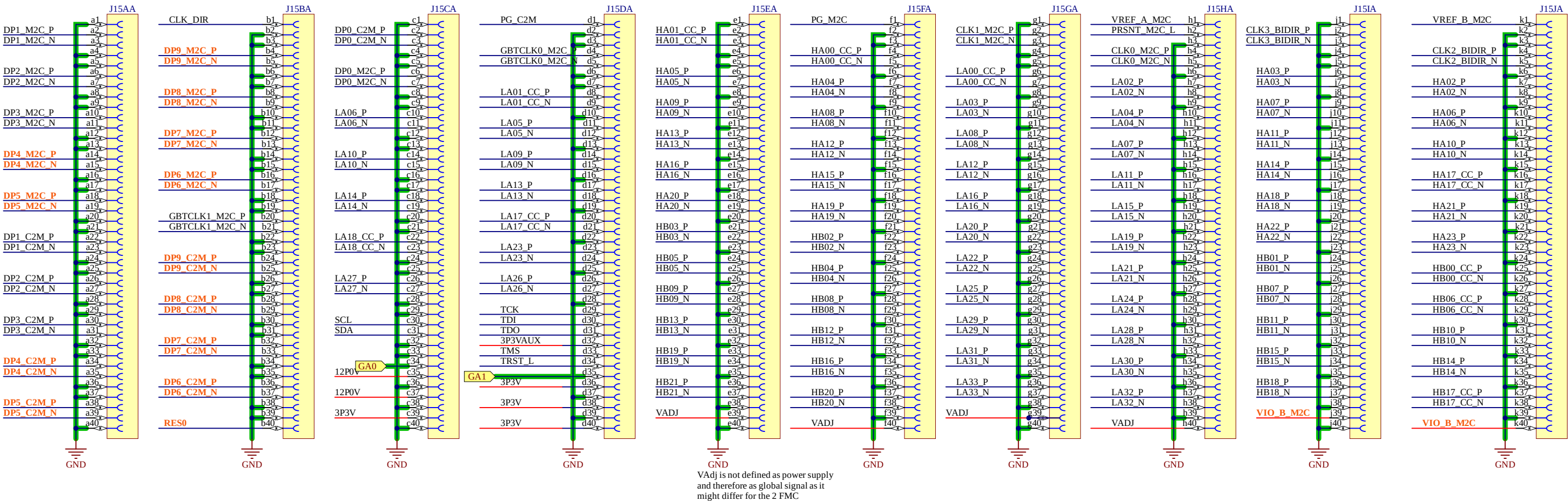
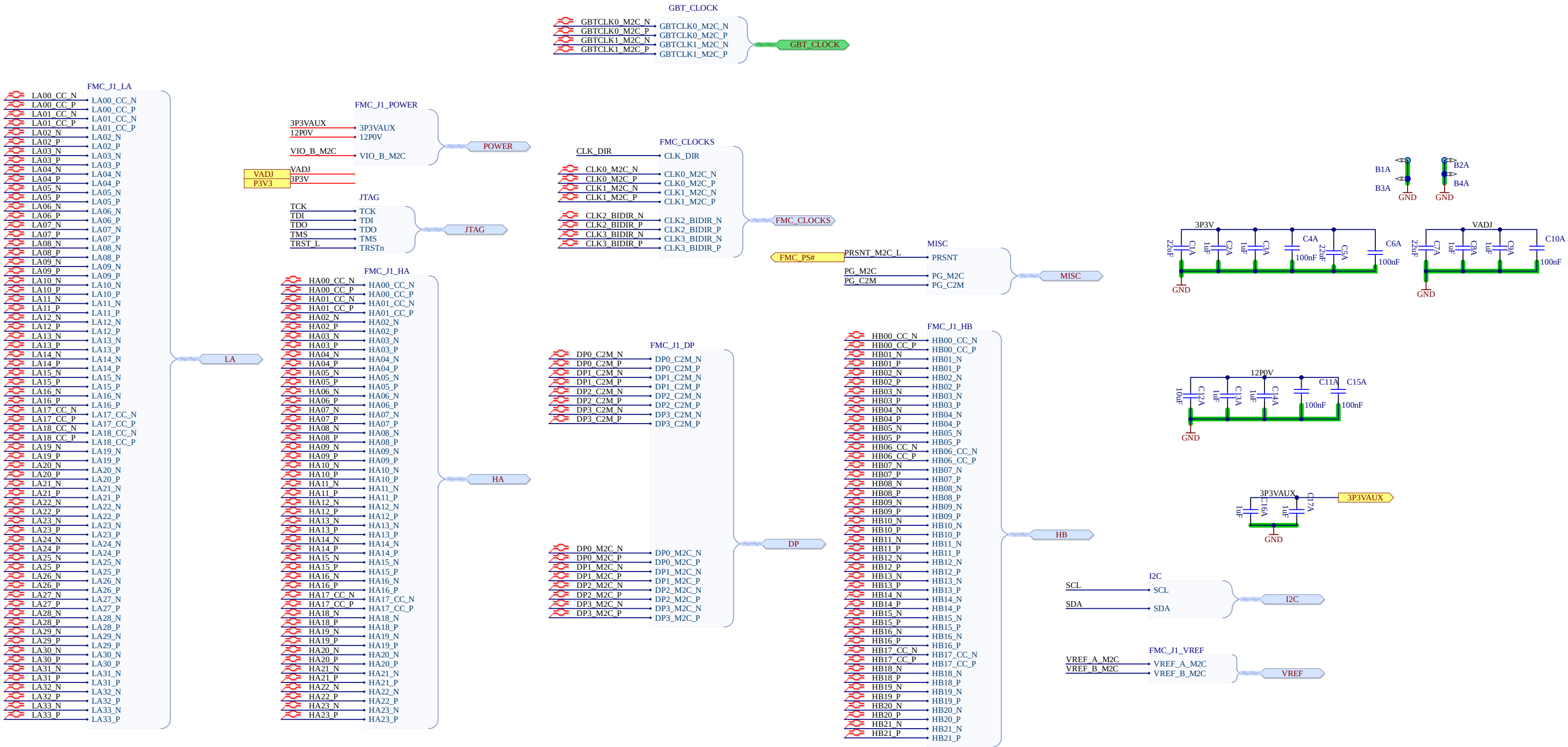
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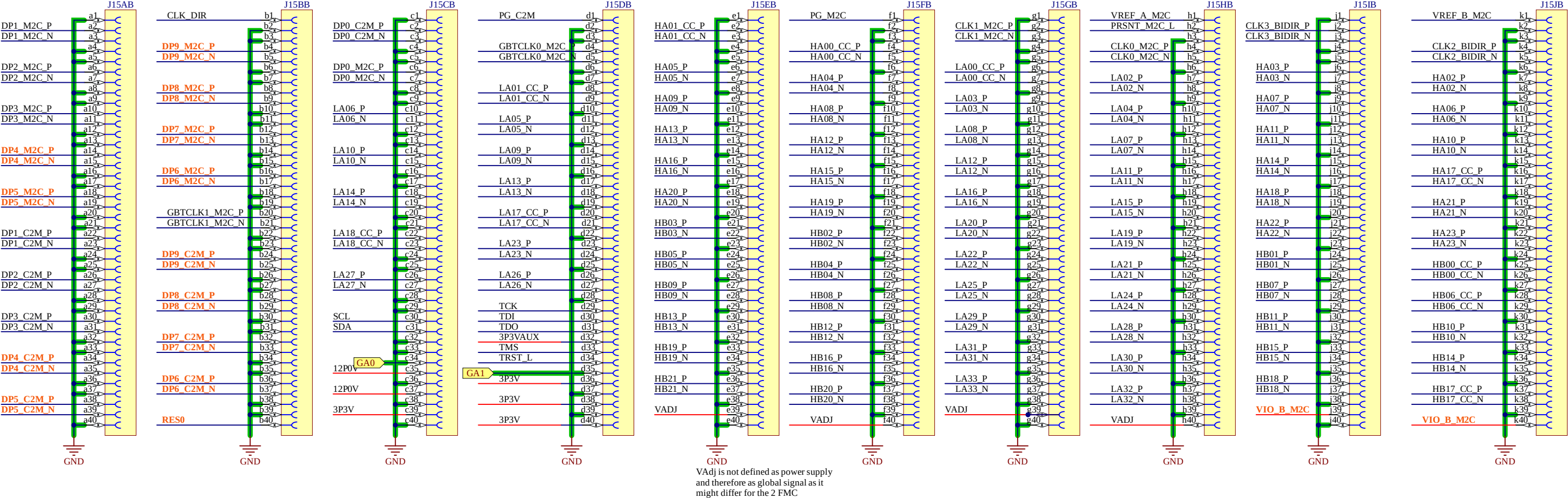
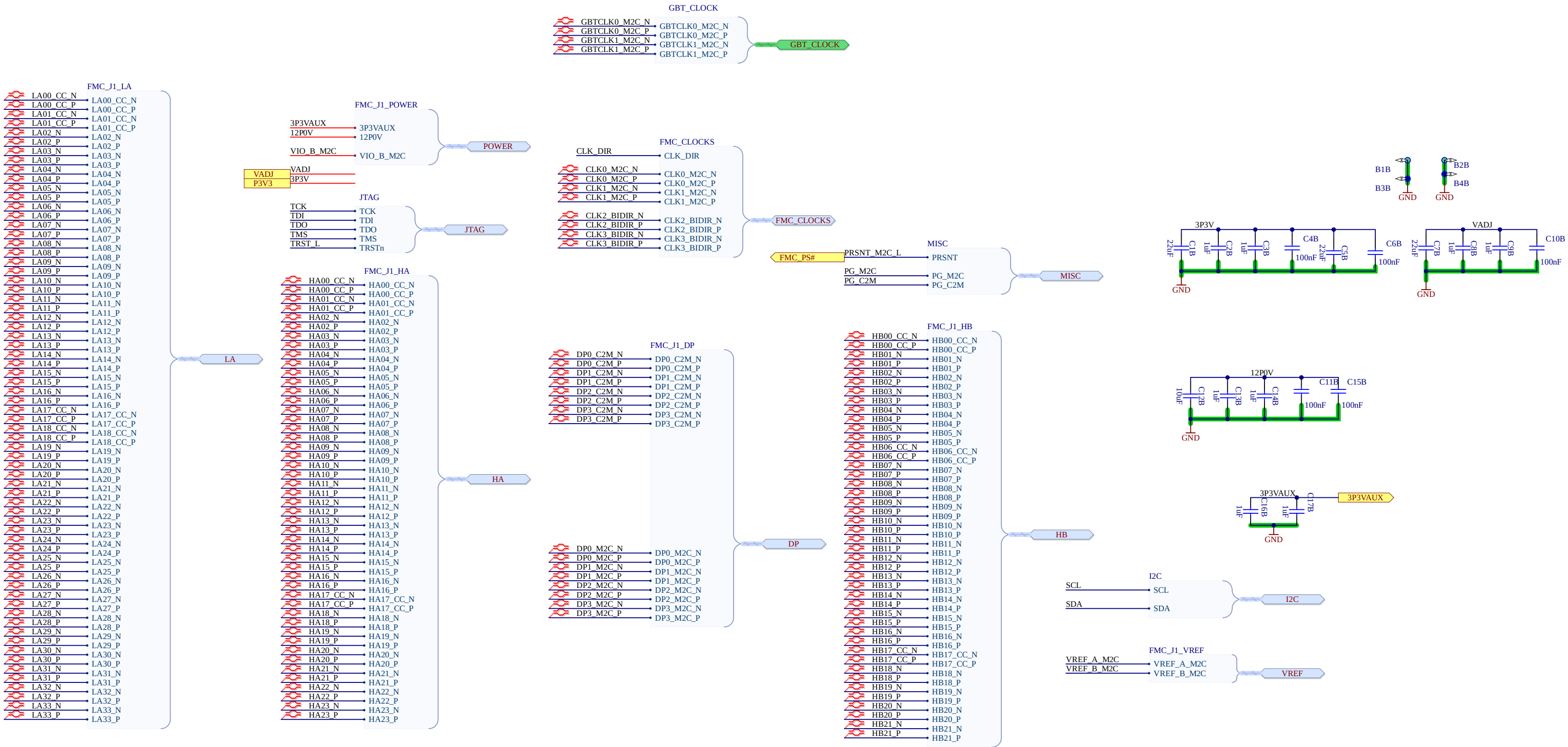
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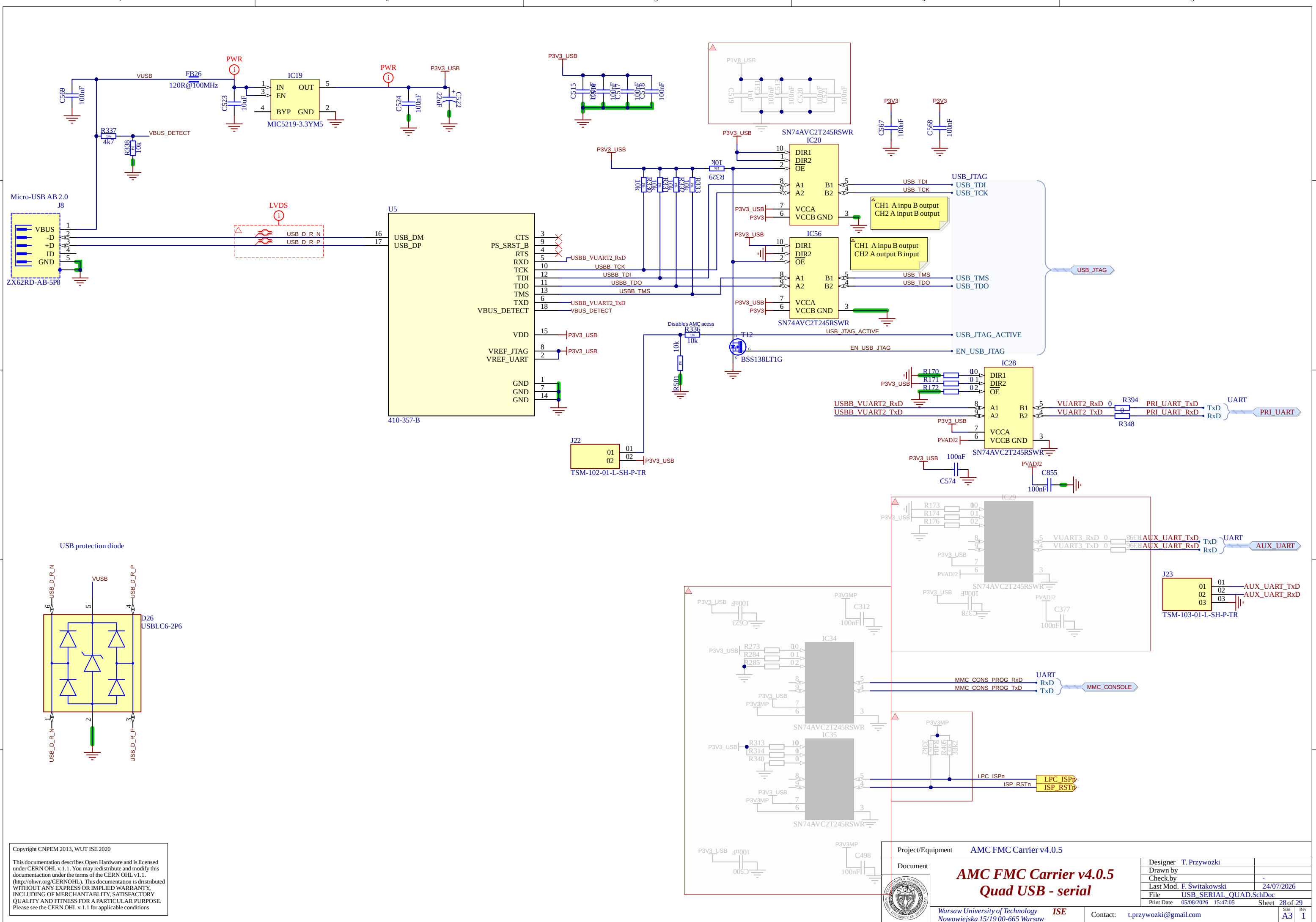
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	Sheet 26 of 29		
	Contact: t.przywozki@gmail.com		Size A4
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VAdj is not defined as power supply and therefore as global signal as it might differ for the 2 FMC





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Warsaw University of Technology ISE		Contact: t.przywozki@gmail.com	
Nowowiejska 15/19 00-665 Warsaw		Size A3	
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A

B

C

D

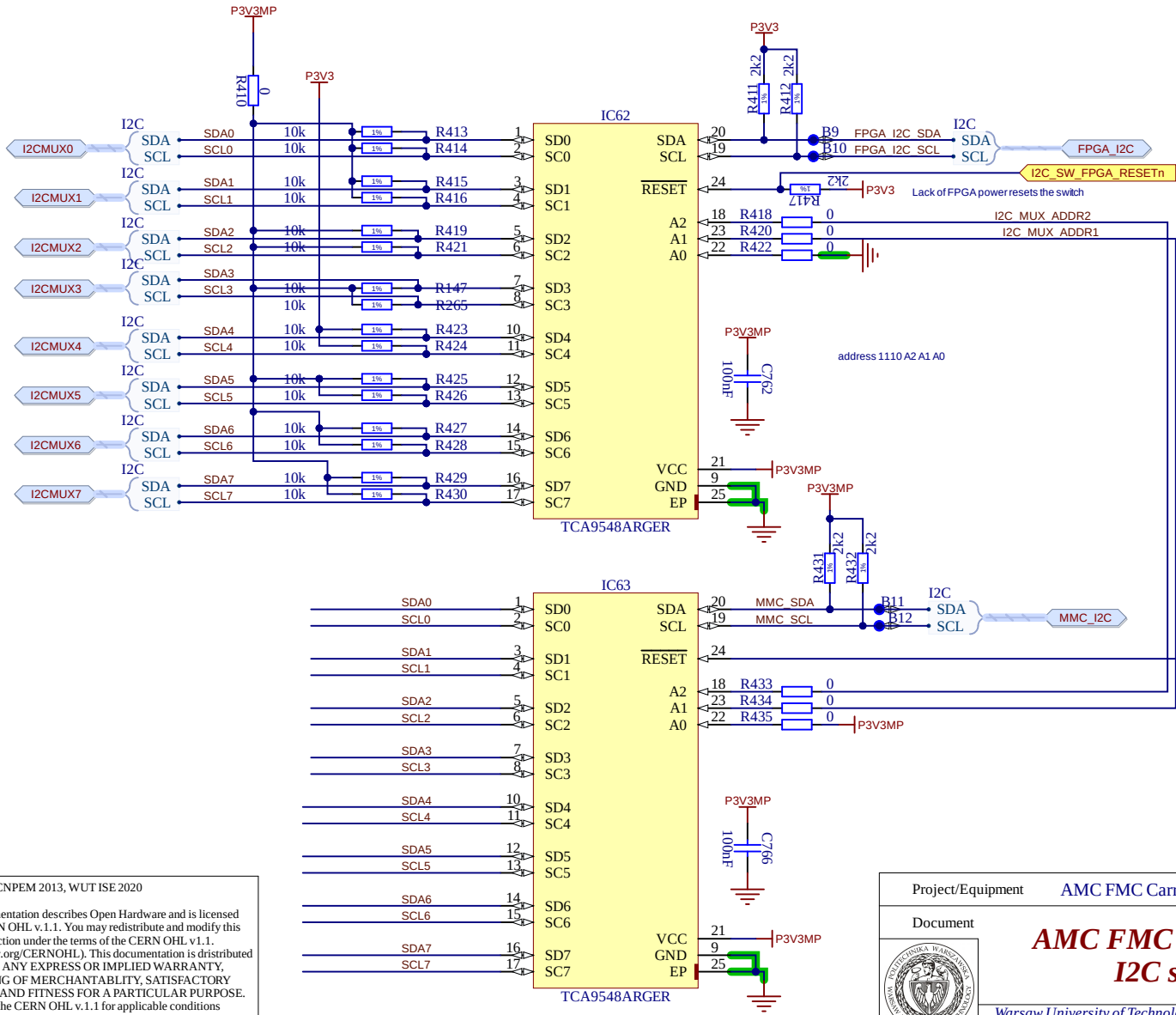
A

B

C

D

I2C switch footprint is compatible with MAX7358 which has interesting anti-lock capabilities



I2C device address map	MUX Port0	MAX6642 0x48
	Temp sensors	LM75: 0x4C LM75: 0x4D LM75: 0x4E LM75: 0x4F
	MUX Port1	RTC EEPROM: 0x57
	RTCE	SRAM/RTCC reg: 0x6F EEPROM 2k: 0x50 EUID: 0x59 EEPROM 64k: 0x51
	MUX Port2	
	MUX Port3	INA3221AIRGV: 0x40
	Power	INA3221AIRGV: 0x41 INA3221AIRGV: 0x42
MUX Port4	Clock switch	8V54816ANLG: 0x5B
MUX Port5	RTM	RTM devices
MUX Port6	FMC2	EEPROM 0x52
MUX Port7	FMC1	EEPROM 0x50

mux address 1110 A2 A1 0 from FPGA side 0x7x
mux address 1110 A2 A1 1 from CPU side 0x7x

I2C_MUX_ADDR[2..1] I2C_MUX_ADDR[2..1]

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Project/Equipment

AMC FMC Carrier v4.0.5

Document

AMC FMC Carrier v4.0.5
I2C switches

Designer

T. Przywozki

Drawn by

Check by

Last Mod.

F. Świtkowski

File

I2C_MUX.SchDoc

Print Date

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Sheet

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Contact:

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Size

A4

Rev

1